

Which border deserves a closer look?

Learning intention

Describe an ecosystem and biodiversity, then use a food web and sample counts to support a cautious recommendation.

Two garden patches each contain 20 counted plants.

Are they equally rich in species?

A garden team needs an evidence brief.

Build the starting vocabulary

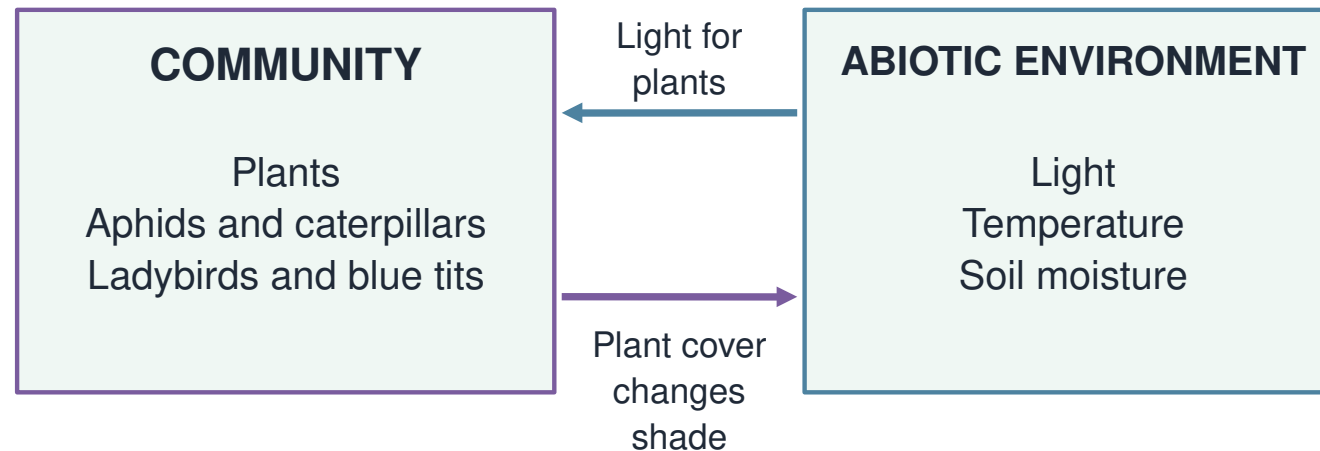
R1. What do plants use light for?

R2. Is soil moisture living or non-living?

R3. Does one sample show every organism at a site?

I explain an interaction

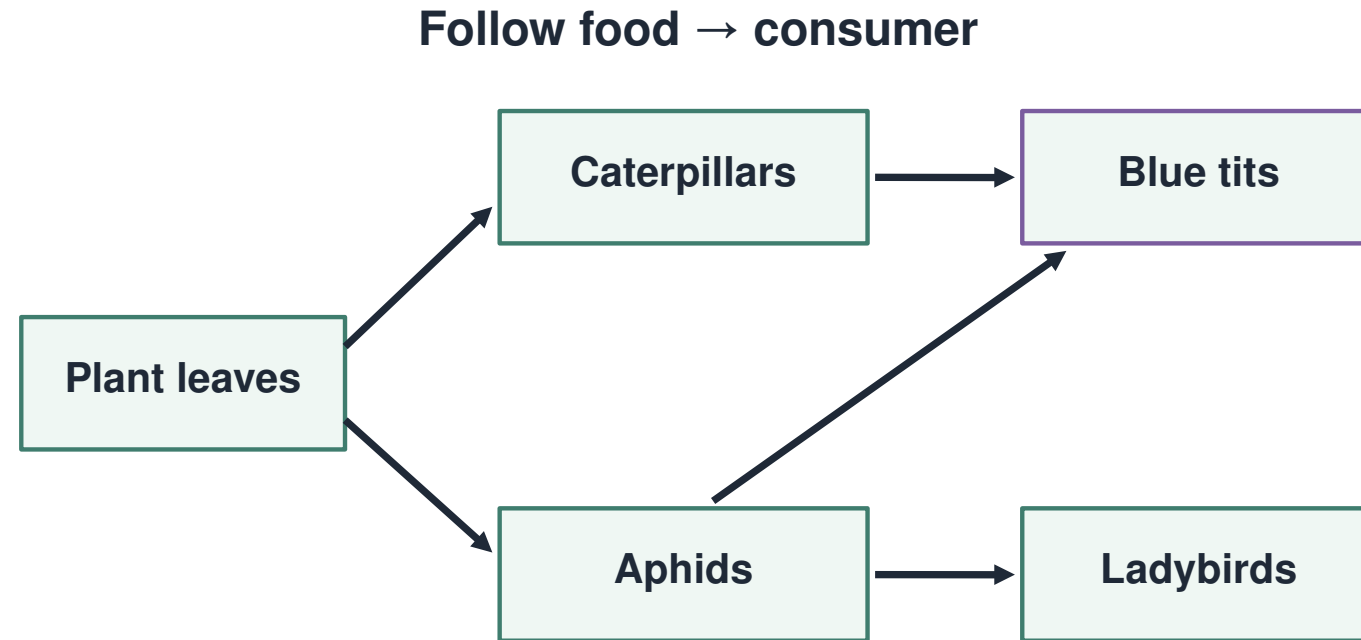
More than a list of living things



Ecosystem = community + environment
+ their interactions.

A community includes populations of different species. An ecosystem includes their environment and interactions. I connect an abiotic factor to a living response.

I read the arrow in the right direction

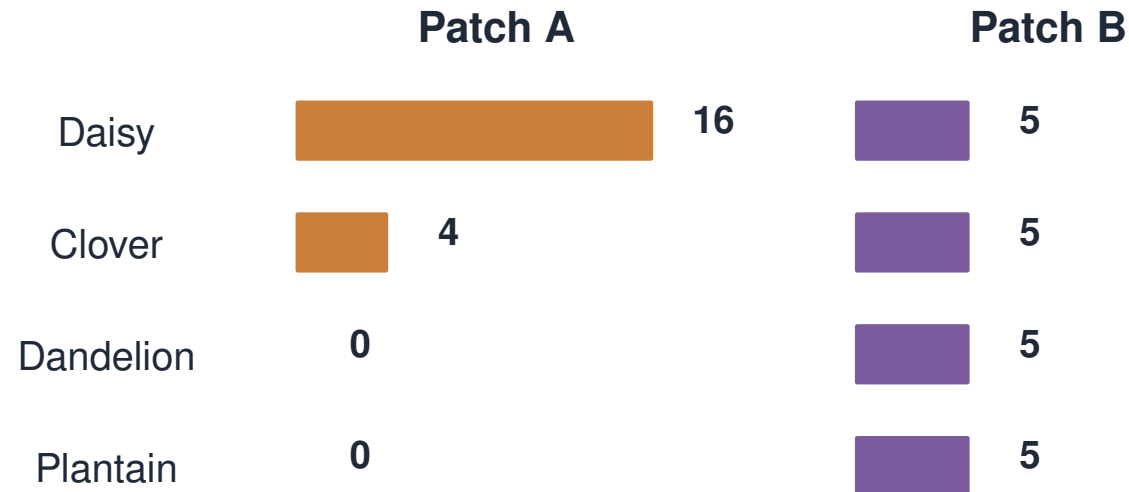


Selected garden feeding links; arrows carry energy and material.

Leaf → aphid means the aphid feeds on the plant. Arrows show food-to-consumer transfer. A web joins several feeding routes.

I count species separately from individuals

Equal totals can hide different richness



Constructed data: one 1 m² sample per patch.
Bar length shows number of individual plants; totals = 20 each.

Abundance counts individuals. Richness counts the different species present. Equal sample totals can have different richness.

The garden evidence desk

W1. Use card A to explain an ecosystem interaction. W2. Trace a possible effect of fewer aphids. W3. Compare counts without overclaiming.

Species	Patch A	Patch B
Daisy	16	5
Clover	4	5
Dandelion	0	5
Plantain	0	5

Constructed data: one 1 m² sample in each patch. Counts are individual plants.

Predict a possible knock-on effect

Aphid abundance falls.

What could happen to ladybirds, and why?

Why is the outcome not guaranteed?

Write a claim the evidence can carry

Both samples contain 20 individuals.

Which has greater recorded richness?

Does this prove a maintenance method caused it?

Choose the defensible headline

Which headline matches our data?

- A** Both patches have identical biodiversity because totals match.
- B** Patch B has greater recorded species richness in these samples.
- C** The mowing method definitely caused the difference.

Write the garden team's brief

Choose one response route.

Explain an interaction and a feeding relationship.

Use a numerical comparison and a limit.

Check your arrows

Can you read each arrow as “is eaten by”?

Does a consumer have another food in the model?

Use “may” when explaining a population change.

Check your evidence language

State what was counted and the sample area.

Separate number of species from total individuals.

Name one next measurement you would make.

Peer-review the brief

Underline the relationship, numerical evidence and limit.

Find one claim that is too strong.

Revise it after feedback.

Give the garden team a question

Choose one next question about the patches.

Explain which evidence would answer it.

We will select a question for later sampling.

Make three ideas precise

E1. What makes an ecosystem more than a community?

E2. What does a food-web arrow show?

E3. Can total abundance alone tell us richness?